Cryptanalysis of Polynomial Learning With Errors (PLWE): A Survey

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Abstract. Lattice-based cryptography (LBC) has emerged as one of the most promising fields supporting post-quantum cryptography (PQC). The Learning With Errors (LWE) problem [18], due to its strong security guarantees, plays a fundamental role in LBC, although it is not very efficient for cryptographic applications. To address this limitation, several variants of LWE have been developed, such as Ring-LWE (RLWE) [15], which is suitable for theoretical purposes, and Polynomial-LWE (PLWE) [20] [7], which is more practical. This survey provides a systematic review of vulnerable instances of PLWE. These attacks may extend to RLWE in instances where the two problems are equivalent. This paper serves as a resource for those seeking a structured overview of the state-of-the-art attacks on PLWE.

Keywords: Polynomial learning with errors · lattice-based cryptography · cryptanalysis

1 Introduction

Over the past decade, significant advances have been made in LBC. Three of the NIST finalists are lattice-based schemes, recognized for their simplicity, strong security guarantees, and applicability to homomorphic encryption [16].

One of the foundational problems in LBC is the Learning With Errors (LWE) problem, introduced by Oded Regev [18]. At a high level, the LWE problem involves recovering a secret $\boldsymbol{s} \in \mathbb{Z}_q^n$ from a system of noisy linear equations $(\boldsymbol{a}, \boldsymbol{b} = \boldsymbol{a} \cdot \boldsymbol{s} + \boldsymbol{e}) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$, where $\boldsymbol{e}$ is sampled from a discrete Gaussian distribution. This problem is believed to be hard even for a quantum machine, due to a reduction from a well-known worst-case lattice problem, the shortest independent vector problem (SIVP) [18]. Despite its strong security guarantees, the presence of a quadratic overhead makes it less efficient for practical applications. To address this, Lyubashevsky *et al.* introduced the ring variant of LWE, the ring learning with errors [15], which offers improved efficiency. RLWE reduces the quadratic overhead while still preserving the security assumptions. Introduced by Stehlé *et al.* [20], the PLWE problem, which is also an algebraic variant of LWE, is preferred for application due to its implementation advantages. However, while this added

structure improves performance, it may also introduce vulnerabilities that attackers could exploit.
This work surveys several known attacks on PLWE and demonstrates how the added structure can be exploited to compromise security. Such a study provides insight into the state of the art on vulnerable instances of this problem, justifying why we should stick to the chosen parameters for cryptographic applications. Moreover, should current standardized schemes be broken and we need alternative, it is important to maintain a catalog of fields and parameters to avoid when constructing new schemes.
For future work, a similar survey could be conducted for attacks on RLWE. An immediate research direction would be to investigate whether existing attacks on RLWE can be extended to PLWE, and vice versa, in cases that have not yet been explored. A natural next step would also be to study the equivalence between PLWE and RLWE over this class of fields.
The remainder of the paper is organized as follows. In Section 2, we review the relevant mathematical concepts and formally define the RLWE and PLWE problems. In Section 3, we describe the known attacks on PLWE, giving a high-level overview of the setup and conditions for success. Furthermore, we analyze these attacks with respect to the parameter choices used in standardized cryptographic schemes.

2 Preliminaries

In this section, we provide the needed background on algebraic number theory and lattice theory necessary for our work. We also recall the definitions of the RLWE and PLWE problems.

Algebraic number theory. A field L is said to be a field extension of K, denoted L/K if $K \subseteq L$. The degree of L/K denoted $[L : K]$ is the dimension of L as a vector space of K. If $[L : K] < \infty$, we say the field extension is finite; otherwise, it is infinite. A number field is a finite field extension of the rationals $\mathbb{Q}$. Given a number field K, if there exists a polynomial $f(x) \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$, for $\alpha \in K$, then α is said to be an algebraic number. If $f(x)$ has integer coefficients, then α is called an algebraic integer. The set of all algebraic integers of the number field K, is called the ring of integers, denoted $\mathcal{O}_K$. A number field K is monogenic if $\mathcal{O}_K = \mathbb{Z}[\alpha]$.
The n-th cyclotomic field is defined by $K = \mathbb{Q}(\zeta_n)$ where ζ_n is a primitive root of unity. The degree of $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is given by $\varphi(n)$, the Euler totient function. Observe that the cyclotomic field is monogenic.
Elements of the number field can be viewed as points in space via the embedding map, which is a structure-preserving injective homomorphism that maps $\sigma : K \to \mathbb{C}^n$. There are exactly n distinct embeddings in a degree n number field $K = \mathbb{Q}(\alpha)$, which maps α to the distinct zeros of the minimal polynomial of α.
The canonical and coefficient embeddings are two important embeddings used in lattice-based constructions. The coefficient embedding which we

denote σ_{coef}, is given by the map,

$$\begin{aligned}\sigma_{coef} &: K \to \mathbb{C}^n \\ \sum_{i=0}^{n-1} a_i x^i &\mapsto (a_0, a_1, \ldots, a_{n-1})\end{aligned}$$

The canonical embedding that we denote σ_{can}, is defined by the map

$$\begin{aligned}\sigma_{can} &: K \to \mathbb{C}^n \\ \alpha &\mapsto (\sigma_1(\alpha), \sigma_2(\alpha), \ldots, \sigma_n(\alpha))\end{aligned}$$

We now recall the definition of the field trace, which maps elements of a number field to its base field $\mathbb{Q}$. This concept will be useful in describing the Attack 3.3

Definition 2.1. *Given a degree n number field, K, and $\alpha \in K$, the trace of α is defined as*

$$Tr_{K/\mathbb{Q}}(\alpha) = \sum_{i=1}^{n} \sigma_i(\alpha)$$

where σ_i are the distinct n embeddings of K.

We recall the following property of the trace function, whose proof can be found in [10].

Lemma 2.2. *Let L/K be a degree n field extension, $\alpha \in L$, and $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in K[x]$ the minimal polynomial of α over K. Then,*

$$\mathrm{Tr}_{K(\alpha)/K}(\alpha) = -a_{n-1}$$

and, in general,

$$\mathrm{Tr}_{L/K}(\alpha) = [L : K(\alpha)]\, \mathrm{Tr}_{K(\alpha)/K}(\alpha)$$

Among the attacks examined below is the smearing attack 3.5. To provide context, we begin by reviewing the following definitions.

Definition 2.3. *(Smearing map)([13]) Suppose $f(x) \in \mathbb{Z}[x]$ is a monic irreducible polynomial of degree n and let α be a root of $f(x) \pmod q$. We define a smearing map as follows:*

$$\begin{aligned}\pi_\alpha &: P_q \to \mathbb{Z}_q \\ g(x) &\mapsto g(\alpha)\end{aligned}$$

Definition 2.4. *([13]) Given a subset $S \subseteq P_q$, we say that S smears under π_α if $\pi_\alpha(S) = \mathbb{Z}_q$.*

Lattices. Given a set of linearly independent vectors $\boldsymbol{b_1}, \boldsymbol{b_2}, \ldots, \boldsymbol{b_m} \in \mathbb{R}^n$, a lattice generated by these vectors is given by

$$\mathcal{L} = \left\{ \sum_{i=1}^{m} a_i \boldsymbol{b_i} : a_i \in \mathbb{Z} \right\}.$$

The rank of the lattice is m, and the dimension is n. When $n = m$, we say that the lattice is full-rank. We will work with full rank lattices. Equivalently, a lattice can also be defined as a pair (L, ρ) where L is a free abelian group of finite rank and ρ, is an injective homomorphism.

Remark 2.5. The pairs $(\mathcal{O}_K, \sigma_{can})$ and $(\mathbb{Z}[x]/(f(x)), \sigma_{coef})$ are lattices and are used to define the RLWE and PLWE problem, respectively.

Probability Distribution Distribution plays an important role in lattice-based protocols, and choosing the right one is crucial for the security, correctness, and efficiency of the cryptographic protocol. The discrete Gaussian distribution has proven to be an excellent choice for the error term, as it enables provable security from a well-known worst-case hard lattice problem while retaining some of the interesting properties of a Gaussian distribution in a lattice setting. For completeness, we begin by recalling the definition of a gaussian distribution.

Definition 2.6. *(Gaussian distribution)*
The Gaussian function over $\mathbb{R}$ with the parameters $\mu = 0$ and $s = \sqrt{2\pi\sigma^2}$ is defined as

$$\rho_s(x) = \exp\left(\frac{-\pi||x||^2}{s^2}\right)$$

for all $x \in \mathbb{R}$. Its corresponding Gaussian distribution, denoted $\mathcal{N}(\mu, \sigma)$, has density function

$$f(x) = \frac{1}{s}\exp\left(\frac{-\pi||x||^2}{s^2}\right)$$

Similarly, in n dimensions, the Gaussian function

$$\rho_s(\mathbf{x}) = \exp\left(\frac{-\pi||\mathbf{x}||^2}{s^2}\right).$$

Since a lattice is discrete, RLWE/PLWE samples are drawn from a discrete Gaussian distribution, which we define below.

Definition 2.7. *(Discrete Gaussian) Given a lattice $\mathcal{L}$, we define the discrete Gaussian distribution $D_{\mathcal{L},s}$ as the probability distribution that assigns the value 0 to all $\mathbf{x} \notin \mathcal{L}$ and the values*

$$D_{\mathcal{L},s}(\mathbf{x}) = \frac{\rho_s(\mathbf{x})}{\rho_s(\mathcal{L})}$$

for every $\mathbf{x} \in \mathcal{L}$. Here $\rho_s(\mathcal{L}) = \sum_{\mathbf{v}\in\mathcal{L}} \rho_s(\mathbf{v})$.

With the necessary background established, we now recall the definition of two structured lattice problems, RLWE and PLWE. In this work, we will focus on the primal version of RLWE which we will simply call RLWE.

Let K be a number field and $R = \mathcal{O}_K$ be its ring of integers. Let $R_q = R/qR$ for a prime q and χ_σ an error distribution that follows a discrete Gaussian distribution over R with standard deviation σ.

Definition 2.8. *(RLWE distribution) The RLWE distribution, denoted $\mathcal{A}_{f,\chi_\sigma,s}$, is defined as follows: uniformly sample $a \in R_q$ at random and $e \leftarrow \chi_\sigma$ and output the pair $(a, b = as + e) \in R_q^2$, where $s \in R_q$ is a fixed secret sampled uniformly at random.*

Definition 2.9. *(Search RLWE) Given access to arbitrarily many samples $(a, b) \in R_q^2$ drawn from the RLWE distribution, the search RLWE problem asks to recover the secret s.*

Definition 2.10. *(Decision RLWE) Given an arbitrary number of samples, the decision RLWE consists of distinguishing between pairs drawn from the RLWE distribution and pairs sampled uniformly at random.*

RLWE enjoys strong security guarantees via a worst-case-to-average-case reduction from a well-known worst-case hard lattice problem. Although it reduces the quadratic overhead associated with LWE, concrete ring structures are still preferred for practical implementation due to their efficiency and the useful properties of their algebraic structures. The PLWE problem, which is of primary interest to our work, addresses this concern, and we define it below.
Let $K = \mathbb{Q}[x]/(f(x))$ be a number field where $f(x)$ is monic and irreducible of degree n and $P = \mathbb{Z}[x]/(f(x))$. Let $P_q = P/qP$ for a prime q and χ_σ an error distribution that follows a discrete Gaussian distribution over P with standard deviation σ.

Definition 2.11. *(PLWE distribution) The PLWE distribution, denoted $\mathcal{B}_{f,\chi_\sigma,s}$, is defined as follows: uniformly sample $a \in P_q$ at random and $e \leftarrow \chi_\sigma$ and output the pair $(a, b = as + e) \in P_q^2$, where $s \in P_q$ is a fixed secret sampled uniformly at random.*

Definition 2.12. *(Search PLWE) Given access to arbitrarily many samples $(a, b) \in P_q^2$ drawn from the PLWE distribution, the search PLWE asks to recover the secret s.*

Definition 2.13. *(Decision PLWE) Given an arbitrary number of samples, the decision PLWE consists of distinguishing between pairs drawn from the PLWE distribution and pairs sampled uniformly at random.*

Unlike RLWE, which enjoys a worst-case to average-case reduction from a well-known hard lattice problem, thus providing a strong security guarantee, no similar reduction is currently known for PLWE. A natural approach to establish a strong security guarantee for PLWE is to establish an equivalence between RLWE and PLWE. Several studies have been conducted in this direction for some special fields [5], [19], [17], [1], [6]. This equivalence is examined by measuring the distortion in the error distribution when mapping PLWE instances to RLWE instances.

Although the additional structures in RLWE and PLWE improve efficiency, they also introduce opportunities for attackers to exploit these added structures. In the next section, we discuss the known weak instances of PLWE.

3 Vulnerable instances of PLWE

In this section, we provide a comprehensive overview of the key results on vulnerable instances of PLWE. We examine their underlying techniques, assumptions, and implications. Throughout this section, we fix the following parameters.

Parameters: For the remainder of the paper, we fix the following parameters. Let $K = \mathbb{Q}[x]/(f(x))$ be a degree N number field with ring of integer $R = \mathcal{O}_K$ and $f(x) \in \mathbb{Z}[x]$ be a monic irreducible polynomial with root α. Set $R_q = R/qR$ and P/qP where q is a prime that splits completely. χ_σ denotes a discrete Gaussian of standard deviation σ.

3.1 Attack when $\alpha = 1$

The first weak instance of PLWE was introduced by Eisenträger, Hallgren, and Lauter in [11]. They showed that, given access to arbitrarily many samples $(a_i, b_i) \in P_q^2$ with condition $f(1) \equiv 0 \pmod q$, it is possible to distinguish PLWE samples from uniform samples. We present a high-level description of the attack algorithm.

1) Given arbitrarily many samples $(a_i(x), b_i(x) = a_i(x)s(x) + e_i(x)) \in P_q^2$
2) Map samples to $\mathbb{F}_q$ via the ring homomorphism
$$f : P_q \longrightarrow \mathbb{F}_q$$
$$(a(x), b(x)) \mapsto (a(1), b(1))$$
3) For each guess of $g = s(1) \in \mathbb{F}_q$, we compute
$$e_i(1) = b_i(1) - a_i(1)g.$$
4) If the correct guess was picked, then $e(1) \sim \chi_{\sqrt{N}\sigma}$. Otherwise, discard and try another guess.

The complexity of the attack is $\mathcal{O}(q)$, which is feasible when q is not too large.

In the same paper, the above attack was extended to any root of small order, which we present below. The order of an element $\alpha \in \mathbb{F}_q$ is the smallest r such that $\alpha^r \equiv 1 \pmod q$. We provide a high-level description of the algorithm below.

3.2 Attack when α has a small order

Suppose r is the order of α a root of $f(x)$. Evaluating at α, we get

$$b(\alpha) - a(\alpha)s(\alpha) = e(\alpha) = \sum_{i=0}^{n-1} e_i\alpha^i = \sum_{j=0}^{r-1}\alpha^j \underbrace{\sum_{i=0}^{\frac{n}{r}-1} e_{ir+j}}_{e_j}$$

We know $e_i \sim \chi_\sigma$. Assume, without loss of generality, that $r \mid n$, then $e_j \sim \chi_{\sqrt{\frac{n}{r}}\sigma}$. Let $\Sigma = \left[\frac{-2n}{r}\sigma, \frac{2n}{r}\sigma\right] \cap \mathbb{Z}$ be the smallness region, with high probability, we have $e(\alpha) \in \Sigma$ and $|\Sigma| \le \left(\frac{4n\sigma}{r}\right)^r$. Unlike the first algorithm, it is harder to distinguish $e(\alpha)$ from uniformly sampled. However, if r is small, we evaluate $e(\alpha)$ for a fixed $s(\alpha) \in \mathbb{F}_q$ and check if it is in the smallness region. If so, return the PLWE sample; otherwise, return the NOT PLWE sample.
The following result gives us the probability of success of this attack.

Proposition 3.1. *([12], proposition 3.1) Assume $|\Sigma| < q$ and let M be the number of samples. If the above attack algorithm 3.2 returns NOT PLWE, then the samples are uniformly distributed. Otherwise, the samples are valid PLWE samples with probability $1 - (|\Sigma|/q)^M$. In particular, this probability tends to 1 as M grows.*

As in the previous attack, this attack is feasible when q is not too large. The attack presented in [11] was later generalized in [12] to include polynomials whose roots are not limited to $\alpha = 1$ or small order, but extend to arbitrary roots. We give a high-level description of the algorithm below.

3.3 Attack when $e(\alpha)$ has small residue.

Introduced in [12], this attack looks at how the error behaves when evaluating at a root α of the defining polynomial modulus q by running a statistical test on $e_i(\alpha)$ and checking if it looks uniform or not. If the samples are randomly generated, we expect $e(\alpha)$ to be uniformly distributed in $\mathbb{F}_q$. And if they cluster around zero, then the guess is correct, and they were PLWE samples.
For a guess g for $s(\alpha)$, let E_i be the event where $b_i(\alpha) - a(\alpha)g \pmod q$ is in the interval $\left[\frac{-q}{4}, \frac{q}{4}\right)$. The objective is to study the probability $P(E_i|D = \chi_\sigma)$ and $P(E_i|D = \mathcal{U})$ where $D = \mathcal{U}$ is the event when (a_i, b_i) is uniformly sampled and $D = \chi_\sigma$ is the event when they are sampled from a PLWE distribution, that is, $b_i(\alpha) - a_i(\alpha)s(\alpha) = e_i(\alpha)$. Let $e_i(\alpha) = \sum_{j=0}^{n-1} e_{ij}\alpha^j$ where $e_{ij} \sim \chi_\sigma$. As we shall see below, this attack also works for the attack 3.1 and 3.2.

- If $\alpha = \pm 1$, then $e_i(\alpha) = \sum_{j=0}^{n-1} e_{ij} \sim \chi_{\sigma\sqrt{n}}$. With high probability, $e(\alpha) \in [-2\sigma\sqrt{n}, 2\sigma\sqrt{n}]$. Observe that if $2\sigma\sqrt{n} < q/4$, then $P(E_i|D = \chi_\sigma) = 1$ for the correct guess, thereby allowing for a distinguishing attack.

- If $\alpha \neq \pm 1$ and α have a small order $r \pmod q$, then

$$e_i(\alpha) = \sum_{i=0}^{r-1} \alpha^j \sum_{i=0}^{\frac{n}{r}-1} e_{ir+j} \sim \chi_{\tilde{\sigma}}$$

where $\tilde{\sigma} = \sigma\sqrt{\frac{n(\alpha^{2r}-1)}{r(\alpha^2-1)}}$. This implies with high probability,

$$e(\alpha) \in \Bigg[-2\sigma\sqrt{\frac{n(\alpha^{2r}-1)}{r(\alpha^2-1)}}, 2\sigma\sqrt{\frac{n(\alpha^{2r}-1)}{r(\alpha^2-1)}}\Bigg].$$

Similarly, if $2\sigma\sqrt{\frac{n(\alpha^{2r}-1)}{r(\alpha^2-1)}} < q/4$, then $P(E_i|D = \chi_\sigma) = 1$ for the correct guess enables a distinguishing attack.

- If $\alpha \neq \pm 1$ and α is not of small order r modulo q, then $e(\alpha) \sim \chi_{\bar{\sigma}}$ where $\bar{\sigma}^2 = \sigma^2 \frac{\alpha^{2n}-1}{\alpha^2-1}$. Hence with high probability, $e(\alpha) \in \big[-2\sigma\sqrt{\frac{\alpha^{2n}-1}{\alpha^2-1}}, 2\sigma\sqrt{\frac{\alpha^{2n}-1}{\alpha^2-1}}\big]$ Analogously, if $2\sigma\sqrt{\frac{\alpha^{2r}-1}{\alpha^2-1}} < q/4$, then $P(E_i|D = \chi_\sigma) = 1$ for the correct guess, enables a distinguishing attack.

Using this setup, we give an overview of the attack algorithm.

- Let l be the number of samples. Pick a guess g for $s(\alpha)$. For each sample, compute
$$e_i(\alpha) = b_i(\alpha) - a_i(\alpha)g.$$
- If there exists a unique guess g mod q such that the event E_i holds for all l samples, return g.
- If more than one guess g satisfies the event E_i for all l samples, return INSUFFICIENT SAMPLES.
- If no guess g satisfies the event E_i for all l samples, return NOT VALID PLWE.

Proposition 3.2. *([12], proposition 3.2) Assume that we are in one of the following cases:*

1. $\alpha = \pm 1$ *and* $8\sigma\sqrt{n} < q$
2. α *has small order* $r \geq 3 \pmod q$, *and* $8\sigma\sqrt{\frac{n(\alpha^{2r}-1}{\alpha^2-1}} < q$.

The algorithm attack 3.3 terminates in time at most $\tilde{\mathcal{O}}(q)$. Furthermore, if the algorithm returns NOT PLWE, then the samples were not valid PLWE samples. If it outputs anything other than NOT PLWE, then the samples are valid PLWE samples with probability at least $1 - (\frac{1}{2})^l$.

Remark 3.3. Set the threshold $N = [\frac{1}{2} + \epsilon l]$ where ϵ is the bias of the distribution. The algorithm is successful with probability

- $P(C < N|D = U) = 1/2$ if $D = U$ and
- $P(C < N|D = \chi_\sigma) > 1/2$ if $D = \chi_\sigma$.

We introduce the next attack, which is an extension of attack 3.2 where the defining polynomial has a quadratic root satisfying certain algebraic properties.

3.4 Attack when $f(x)$ has a quadratic root whose trace is zero.

Building on the attack 3.2 presented above, the method introduced in [3] exploits the number-theoretic properties of the trace function to achieve a small-smallness region without requiring the order of the root to be small.

Setup: Suppose $f(x)$ has a quadratic factor of the form $x^2 + \rho$ where $\rho \in \mathbb{F}_q[x]$ has multiplicative order r. Then $\exists \alpha \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$ such that $\alpha^2 = -\rho$ and $f(\alpha) = 0$. Observe that $\mathrm{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(\alpha) = 0$.
Given a PLWE sample $(a(x), b(x) = a(x)s(x) + e(x)) \in R_q^2$, we compute the trace

$$\mathrm{Tr}(e(\alpha)) = \mathrm{Tr}(b(\alpha) - a(\alpha)s) = 2\sum_{k=0}^{r-1}(-\rho)^k \sum_{i=0}^{\frac{N}{r}} e_{2(ir+k)}, \qquad (1)$$

where $N := \lfloor \frac{n-1}{2} \rfloor$. With high probability, the size of the set of possible values for the $\mathrm{Tr}(e(\alpha))$ is bounded above by $|\Sigma| \leq \left(2\sqrt{\frac{N}{r}}\sigma\right)^r$.
Whereas the previous attack 3.2 relies on the multiplicative order r being small, the authors of [3] showed that this requirement can be removed by instead ensuring that the bound on the smallness region is sufficiently small. The attack algorithm is described below.

Attack algorithm: This attack is carried out in two stages. First, the authors attack the PLWE problem over a smaller ring $R_{q,0} \times R_q$ where $R_{q,0} = \{p(x) \in R_q : p(\alpha) \in \mathbb{F}_q\}$ is an $\mathbb{F}_q$-vector subspace of R_q of dimension $N-1$. In the second stage, they demonstrate that solving PLWE on the larger ring R_q^2, can be reduced to solving PLWE samples over the smaller ring, $R_{q,0} \times R_q$ under certain defined conditions. We describe the attack algorithm on the smaller ring below.
Attack on the smaller ring $R_{q,0} \times R_q$: Given a PLWE sample $(a_i(x), b_i(x)) \in R_{q,0} \times R_q$, pick a guess $g = s(\alpha) \in \mathbb{F}_{q^2}$ and check if

$$\mathrm{Tr}(e(\alpha)) := \frac{1}{2}\mathrm{Tr}(b_i(\alpha) - a_i(\alpha)g) \in \Sigma.$$

If yes, keep s and try the next. If not, safely remove from $\mathbb{F}_{q^2}$ not only s, but also all elements $s' \in \mathbb{F}_{q^2}$ such that $\mathrm{Tr}(s') = \mathrm{Tr}(s)$ (by Theorem 2.2). So we can remove q elements from S at a time. This reduces the size of the set from which you select your guesses.
If there is only one guess g, then the samples are PLWE samples. If no valid guess g is found, then the samples are uniformly random. If more than one guess is valid, add more samples, and rerun the algorithm. The complexity of this attack for M samples is $\mathcal{O}(Mq)$.
Although it is weaker than the algorithm in 3.2, it is effective as a decision attack. The following result gives us the success probability.

Proposition 3.4. *([3], proposition 3.7) Assume that $|\Sigma| < q$. If the attack Algorithm on the smaller ring $R_{q,0} \times R_q$ returns **NOT PLWE**, then the samples come from the uniform distribution on $R_{q,0} \times R_q$.*

If it outputs anything else than ***NOT PLWE****, then the samples are valid PLWE samples with probability* $1 - (|\Sigma|/q)^M$*. In particular, this probability tends to 1 as* M *grows.*

For the second stage of the attack, we assume we have access to an algorithm that can be modeled as a random variable X_0. This algorithm samples pairs $a(x), b(x)) \in R_q^2$ distributed according to a random variable X, and returns a pair $(a(x), b(x)) \in R_{q,0} \times R_q$ along with the number of occurrences. The following result then holds.

Proposition 3.5. *([3],proposition* 3.8*).*
- *If* X *is uniformly over* R_q^2*, then* X_0 *is also uniform over* $R_{q,0} \times R_q$*.*
- *If* X *follows a PLWE distribution then* X_0 *is an* $R_{q,0} \times R_q$ *valued PLWE distribution.*

The probability that given l samples from R_q^2, k of them will be in $R_{q,0} \times R_q$, is around $\frac{1}{2}$ if the condition $k \approx lq^{-1} > 5$ is satisfied.

A generalization of this attack was presented in [9] where instead of $f(x)$ having a quadratic root with zero trace, the condition was extended to $f(x)$ having a root of the form $x^n + \rho$ with trace zero. The attack is constructed analogously to attack 3.4 where an attack is designed on a smaller ring $R_{q,0} \times R_q$ and then pulled back to the larger ring.
Using a similar strategy of exploiting roots with trace zero, the authors of [4] designed an attack on PLWE over a subring $R_{q,0} \times R_q$ of a cyclotomic field, where the integer modulus q does not split completely. However, unlike in the attack 3.4, the authors were unable to extend this attack to the larger ring. Studying such an extension remains an interesting open problem, as it will demonstrate that the requirement for q to split completely in cryptographic applications is motivated not only by efficiency but also by security.

3.5 Attack when the smearing condition holds

Proposed as an open problem in [11] and subsequently addressed in [2], the smearing attack provides a method to reduce PLWE samples in P_q to samples in $\mathbb{F}_q$ and analyze the behavior of the error to distinguish them from uniform samples. Before stating the result that justifies the use of the smearing condition as a distinguishing attack, we set some parameters.

Parameters: Let N be the number of trials (assumed odd), and m be the number of samples per trial. Define $P(m, q, U)$ as the probability that m samples were drawn independently from U smears and $P(m, q, \chi)$ as the probability that m samples were drawn independently from χ, smear. Let $[q] = \{1, 2, \cdots, q\}$

Theorem 3.6 (5.2). *Let* χ *be a probability distribution over* $[q]$ *and let* U *be the uniform distribution over* $[q]$*. Then,* $P(m, q, \chi) \leq P(m, q, U)$*, with equality iff* $\chi = U$*.*

We give a high-level setup of the attack.
Setup:
- Choose m such that
  - $P(m, q, U) > 1/2$ (by Theorem 3.6)
  - $P(m, q, \chi) < 1/2$
- In each of the N trials, take m samples and check if they smear.
- If more than half of the trials result in smearing, we conclude samples are from a uniform distribution. Otherwise, conclude samples came from a non-uniform distribution.

Since theorem 3.6 shows that smearing is more likely to occur if samples are uniformly distributed, the setup accurately identifies the source of the data with high probability. We now describe the smearing attack algorithm.

The smearing attack algorithm Given access to many samples $(a_i, b_i) \in P_q \times P_q$, test if the samples are PLWE samples or uniform samples.
- Choose m and N such that type 1 error α (false positive i.e the classified as PLWE and false negative (PLWE classified as uniform)
- For each guess $g = s(\alpha) \in \mathbb{F}_q$ and for N independent trials, draw m samples $(a_i(x), b_i(x))$ for N independent trials and compute $S = \{b_i(\alpha) - g a_i(\alpha)\}$ and then test if $S \subset \mathbb{F}_q$ smears.
- Let C be the number of trials that result in smearing.
  - - If $C > 1/2$, then the error distribution is uniform
  - - Otherwise, it is Gaussian.

Decision
- If for all $g \in \mathbb{F}_q$, $e(\alpha)$ it looks uniform, then the samples came from a uniform distribution.
- If for only one $g \in \mathbb{F}_q$, $e(\alpha)$ is non-uniform, then the samples were PLWE samples. In fact, g is likely the right guess for $s(\alpha)$ with high probability.
- If for more than one guess $g \in \mathbb{F}_q$, $e(\alpha)$ looks non-uniform, then retry with new parameters m and N.

With the right choice of parameters, the probability of success is high. We justify this claim with the following result from [2].

Theorem 3.7. *([2], proposition 5.2) Let α and β be, respectively, the type 1 and type 2 errors of the smearing decision on q. Then, if the true distribution of the samples (a_i, b_i) is the uniform distribution over $P_q \times P_q$, the smearing attack gives the correct decision with probability*

$$\frac{1-\alpha}{1+(q-1)\alpha}$$

If the true distribution of the samples (a_i, b_i) is the PLWE distribution over $P_q \times P_q$, the smearing attack gives the correct decision with probability

$$\frac{1-\alpha-\beta+q\alpha\beta}{1-\alpha+(q-1)\alpha\beta}$$

Extending attacks on PLWE to RLWE. An interesting question arises: can some (or all) of the attacks listed above on PLWE be extended to RLWE? Research in this direction has answered this question in the affirmative for some of the attacks. One of the main approaches involves transforming RLWE samples into PLWE samples and then applying the corresponding PLWE attack. This transformation introduces some distortion, which is measured by the spectral norm ρ of the transformation matrix.

In [12], the authors demonstrate how the attack described in 3.1 can be applied to the decision RLWE problem. For this attack to succeed, the following conditions must be satisfied:

- the number field K is monogenic
- the defining polynomial $f(x)$ satisfies the equation $f(1) \equiv 0(\pmod{q})$, and
- the spectral norm ρ and error width σ are sufficiently small.

Building on this, the authors of [8] extended the result to the search RLWE problem achieving 100% success probability using fewer samples. While these various attacks raise important concerns, they do not imply that the NIST-standardized schemes have been broken. We elaborate on this in the following section.

3.6 Impact of These Attacks on Practical Parameter Choices

The attacks on PLWE discussed above do not pose a threat to cryptosystems based on this problem. For both efficiency and security, the ring used is typically $R = \mathbb{Z}[x]/f(x)$. The defining polynomial $f(x)$ is chosen to be the n-th cyclotomic polynomial where n is a power-of-two, *i.e.,* $n = 2^k$, where $k \in \mathbb{Z}$. The modulus q is selected such that $q \equiv 1 \pmod{n}$. Additionally, for homomorphic computations, the size of q should be at least $q = 2^{128}$ [14]. We discuss this claim below.

- Attack 3.1 fails in this setting because $\alpha = 1$ is not a root of $f(x)$ when $q \nmid n$. More so, although the attack complexity is linear in q, it remains infeasible in practice since q is large.
- Attack 3.2 fails in this setting due to the fact that $q \equiv 1 \pmod{n}$, which implies that all the roots of $f(x)$ have order n. As a result, there are no roots of small order.
- Attack 3.3 fails in this setting because, although its complexity is linear in

  q, it is infeasible in practice due to the large size of q.
- Attack 3.4 fails in this setting because it is nearly impossible to find a cyclotomic polynomial with a quadratic factor of the form $x^2 + \rho$.
- Attack 3.5 fails in this setting because, although its complexity is linear in q, it is infeasible in practice due to the large size of q.

3.7 Conclusion

In this survey, we have described some of the attacks on PLWE that exploit the algebraic structures of the polynomial ring. For practical constructions, these conditions are carefully avoided. In particular, the use of power-of-two cyclotomic polynomials ensures that such weak instances do not arise. However, these attacks provide a useful catalog of parameters to avoid if cyclotomic polynomials become inefficient and insecure, and we need to construct new schemes.

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